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Batch → Ramanujan
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Q-1

(a) Yes there is 1 type of bias →

Selection Bias → It is because only students present in the library are surveyed. These students are more likely to have better study habits than those outside the library.

(b) Better sampling technique to obtain more representative sample is Simple Random Sampling or Stratified Sampling, where every student in the college has an equal chance of being selected.

Q-2 Given $\lambda \to 0.01$

(a) mean time of failure for exponential distribution is →

$$\text{mean} = \frac{1}{\lambda}$$

$$\therefore E[T] = \frac{1}{\lambda}$$

$$= \frac{1}{0.01} = 100 \text{ hours}$$

$\boxed{\text{So, mean} = 100 \text{ hours}}$

(b) $P(T > 200) = ?$

$P(T > t) = e^{-\lambda t}$ (Survival function)

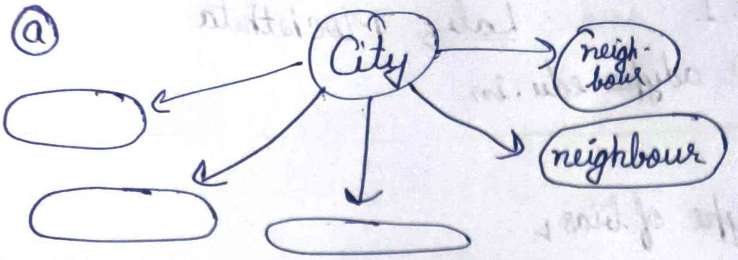
$$= e^{-0.01 \times 200} = e^{-2}$$

$$\simeq 0.1315$$

$$\boxed{P(T > 200) = 0.1315}$$

Q-3

ⓐ



Select 5 group out of all neighbours

Method Used → Cluster Sampling

ⓑ

Cluster Sampling	Simple Random Sampling
• Select groups	• Select individual
• faster	• more accurate
• Less precise	• more representative

ⓒ

Advantages	Drawbacks
• Cost Effective	• Less Accurate
• Easy to implement	• High Sampling Error.

Q-4 total pens = 8
choose 2 ⟶ total outcomes → $^8C_2 = \frac{8!}{6! \times 2!} = 28$

ⓐ Joint PMF

(X, Y)	red	blue	green	ways	probability
(0,0)	0	0		$^3C_2$	3/28
(0,1)	1	0	1	6	6/28
(0,2)	2	0	0	1	1/28
(1,0)	0	1	1	9	9/28
(1,1)	1	1	0	6	6/28
(2,0)	2	0	0	3	3/28

(b) $P[(x,y) \in A]$ where $\{(x,y) \mid x+y <= 1\}$

Valid Cases

$(0,0), (1,0), (0,1)$

$$P = \frac{3+9+6}{28} = \frac{9}{14}$$

$$P[x,y) \in A] = 9/14$$

Q→5 $P(x,y) = k(2x+y)$

(a) Sum of all probabilities equals to 1.

$X = \{0,1,2\}$

$Y = \{0,1\}$

$\sum k(2x+y) = k\sum(2x+y) = 1$

$0+1+2+3+4+5 = 15$

$15(k) = 1 \Rightarrow k = \frac{1}{15}$

(b) Marginals:

$$P_X(x) = \sum_y p(x,y)$$

$x \to 0 \to 1/15$

$x \to 1 \to 5/15 = \frac{1}{3}$

$x \to 2 \to 9/15 = \frac{3}{5}$

$P_Y(y)$

$y=0 \to \frac{6}{15}$

$y=1 \to 9/15$

(c) $P(X=x \mid Y=1)$

$$P(X=x \mid y=1) = \frac{P(x,1)}{P(Y=1)}$$

$P(Y=1) = 9/15$

$x=0 \rightarrow \frac{1}{9}$

$x=1 \rightarrow 3/9$

$x=2 \rightarrow 5/9$

(d) $E[X]$ & $E[Y]$

$E[X] = x \cdot P(x)$

$= 0 \times \frac{1}{15} + 1 \times \frac{3}{15} + 2 \times \frac{5}{15}$

$P_X(0) = P(0,0) + P(0,1) = \frac{1}{15}(0+1) = \frac{1}{15}$

$P_X(1) = \frac{1}{15}(2+3) = \frac{5}{15}$

$P_X(2) = \frac{1}{15}(4+5) = \frac{9}{15}$

$\boxed{E[X] = \frac{23}{15}}$

$E[Y]$

$P_Y(0) = \frac{0+2+4}{15} = \frac{6}{15}$

$E[Y] = 0 \times \frac{6}{15} + 1 \times \frac{9}{15} = \frac{9}{15}$

$P_Y(1) = \frac{1+3+5}{15} = \frac{9}{15}$

$\boxed{E[Y] = \frac{9}{15}}$

(e) $Cov(X,Y) = E[XY] - E[X]E[Y]$

X	Y	$P(X,Y)$	$xy \cdot P(X,Y)$
0	0	0	0
0	1	1/15	0
1	0	2/15	0
1	1	3/15	3/15
2	0	4/15	0
2	1	5/15	10/15

$E(xy) = \frac{13}{15}$

$\frac{13}{15} - \frac{23}{15} \times \frac{9}{15} = \frac{195-207}{225} = -0.0533$

Ⓒ

(f) It is Not Independent

since $P(x,y) \neq P(x).P(Y)$

Q-6

(a) Stratified

(b) Systematic

(c) Cluster

(d) Simple Random

(e) Stratified

Q-7

$\mu = 0.2 \qquad \sigma = 0.1 \qquad n = 50$

$$\sigma_x = \frac{\sigma}{\sqrt{n}} = \frac{0.1}{\sqrt{50}} = 0.01414$$

$$z = \frac{0.24 - 0.2}{0.01414} = 2.83$$

$$P(Z > 2.83) = 0.0023$$

Q-8

x	2	3	5	7	9	11	13	15
y	4	6	7	10	9	13	12	15

(a) $\bar{X} = \frac{2+3+5+7+9+11+13+15}{8} = 8.125$

$\bar{Y} = \frac{4+6+7+10+9+13+12+15}{8} = 9.5$

$Cov(x,y) = \frac{1}{n}\left(\Sigma (x-\bar{x})(y-\bar{y})\right) = \frac{114.5}{8}$

$$Cov(x,y) = 14.3125$$

(b) Coorelation (r)

$$r = \frac{Cov(x,y)}{\sigma_x \, \sigma_y}$$

$$\sigma_x = \sqrt{\frac{1}{8}\left(\Sigma(x-\bar{x})\right)^2} = 4.622$$

$$\sigma_y = \sqrt{\frac{1}{8}\left(\Sigma(y-\bar{y})\right)^2} = 3.354$$

$$r = \frac{14.3125}{4.622 \times 3.354} = 0.924$$

(c) $r = 0.923$, <u>Strong Positive</u>

as $x \uparrow$ so $y \uparrow$

Q-9 <u>Given</u> mean $\mu = 3\,cm$

$\sigma = 0.05\,cm$

specification 3.01 0.01 = [2.99 3.01]

we ~~m~~ need

$$P(\ ) = P(X < 2.99) + P(X > 301)$$

$$Z = \frac{X-\mu}{\sigma}$$

$x = 3.01$ | $X = 2.99$

$$z = \frac{0.01}{0.005} = 2 \qquad | \qquad Z = -2$$

$$P(\text{Scrapped}) = P(Z < -2) + P(Z > 2)$$

$$= 2 \times 0.0228$$

$$= 0.0456$$

<u>about</u> 4.56% of ball bearing were rejected.

(c) $\boxed{Q-10}$

(a) $x \sim U(0, 10)$ (Given)

for uniform distribution

$$P(a \le x \le b) = \frac{b-a}{10-0} = \frac{b-a}{10}$$

max waiting time is 10min.

So, probability of waiting more than 10min is 0

(b) $P(2 \le x \le 7) = \frac{7-2}{10-0} = \frac{5}{10} = \frac{1}{2}$

$$P(2 \le x \le 7) = \frac{1}{10}$$

$\boxed{Q-11}$ Total Population = 600

minimum = 95

mean = ~~ttt~~ μ = ~~95~~ 115

standard deviation $\sigma = 12$

$P(x < 95) = ?$

$$z = \frac{95-115}{12} = \frac{-20}{12} = -1.67$$

$$P(z < -1.67) = 0.0475$$

Rejects students = $600 \times 0.0475 = 28.5$

$\therefore$ 29 students were rejected.

$\boxed{Q-12}$ (a) $P(T > 150 / T > 100)$

$$P(T > 150 / T > 100) = \frac{P(T > 150 \cap T > 100)}{P(T > 100)}$$

$$= \frac{P(T > 150)}{P(T > 100)} = \frac{e^{-\lambda T_1}}{e^{-\lambda T_2}} = \frac{e^{-\lambda 150}}{e^{-\lambda 100}}$$

$$= e^{-50\lambda}$$

(b)

$P(T > 50) = e^{-50\lambda}$

(c) This result shows that, the probability of bulb lasts more than 50 hrs, git given that it has already lasted 100 hrs.

Q-13

(a) All person in the city of Richmond who are eligible to vote in the school board election.

(b) All possible outcomes of tossing this coin under the same conditions

(c) All pairs of this new tennis shoe that could be used under similar player condition.

(d) All posible travel timer for the Sawyer Communicating from her suburban home to her mid town office.

Q-14

(a)

1) Selection bias
2) Time bias
3) Lifestyle bias
4) Undercoverage bias
5) Response bias.

(b) The estimate of 6 hr perday is not reliable because,:

1. The sample is not selected randomly.
2. It represents only a specific subgroup.
3. Important segments of the student population is underpresent or excluded.
4. Therefore the result may not reflects the true.

(c)

1. Simple Random
2. Stratified
3. Sampling at different time or location

Q-15

$\mu = 28$ $\quad n = 40$ $\quad \sigma = 5$

$P(\bar{x} > 30)$

By CLT, $\bar{X} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$

standard deviation

$$\frac{\sigma}{\sqrt{n}} = \frac{50}{\sqrt{40}} = 0.79$$

$$Z = \frac{30-28}{0.79} \simeq 2.53$$

From table, $P(Z > 2.53) \simeq 0.0057$

Q-16

(a) $\mu = 99.61$ $\quad \sigma = 0.08$

$$Z_1 = \frac{99.5 - 99.61}{0.08} = \frac{-0.11}{0.08} = -1.375$$

$$Z_2 = \frac{99.3 - 9.61}{0.08} = 1.13$$

Distribution: Normal.

$P(Z < -1.38) = 0.0838$

$P(99.5 < x < 99.7) = 0.708 - 00838$

$= 0.787$

(b) we need x such that

$P(X > x) = 0.05$

This corresponds to $P(Z < 2) = 0.95$

From Z table,

$z = 1.645$

now, $x = \mu + z\sigma = 99.61 + (1.645)(0.8)$

$x = 99.61 + 0.1316$

$= 99.7416$

$\boxed{Q-17}$

$\sigma = 5.8 kg$

$\mu = 78.3 kg$

$n_1 = 64 \quad n_2 = 100$

$Var(\bar{x}) = \frac{\sigma^2}{n}, \quad \sigma^2 = (5.6)^2 = 31.36$

for $n = 64$

$Var(\bar{x}) = \frac{31.36}{64}$

$= 0.49$

for $n = 100$

$Var(\bar{x}) = \frac{31.36}{100}$

$= 0.3136$

Reduction in Variance $= 0.1764$

$\boxed{Q-18}$

ⓐ Given: $X_1, X_2, ---- X_n$ are i,i,d with $E[X_i] = \mu$

$Var(X) = \sigma$

for $T_i = \bar{X}_1$

$E[T_i] = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n}\left(\sum E[X_i]\right) = \frac{1}{n} \times n\mu = \mu$

T_1 is unbiased.

$T_2 = \frac{X_1 + X_2}{2}$

$E[T_2] = \frac{1}{2}(\mu + \mu) = \mu$

T_2 is unbiased.

(b) Var of T_1 = $Var(X) = \frac{\sigma^2}{n}$

Var of T_2 = $Var\left(\frac{X_1+X_2}{2}\right) = \frac{1}{4}\left(Var(X_1) + Var(X_2)\right)$

$= \frac{1}{4}(\sigma^2+\sigma^2) = \frac{\sigma^2}{2}$

for $n > 2$

$\frac{\sigma^2}{n} < \frac{\sigma^2}{2}$

(c) Both are unbiased but T_1 has smaller variance. So, T_1 is prefferable.

Q-19

(a) $X_i \sim N(0, i^2)$. So, $E[X_i] = 0$ $Var(X_i) = i^2$

$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \longrightarrow$ Sample Mean

$Var(\bar{x}) = Var\left(\frac{1}{n}\sum_{1}^{n} X_i\right) = \frac{1}{n^2}\sum_{1}^{n} Var(X_i)$

but $Var(X_i) = i^2$

$Var(\bar{X}) = \frac{1}{n^2} \cdot \sum_{1}^{n}(i^2)$

$Var(\bar{X}) = \frac{(n+1)(2n+1)}{6n}$

(b) The law of larger numbers states that As the sample size tends $n \to \infty$, the sample mean $\bar{X}_1$ converts to the population mean.

Mathematically,

$\bar{X}_n \to \mu$

Interpretation,

1. When we take more ~~than~~ observation
2. The avg of the sample get closer to the true mean.